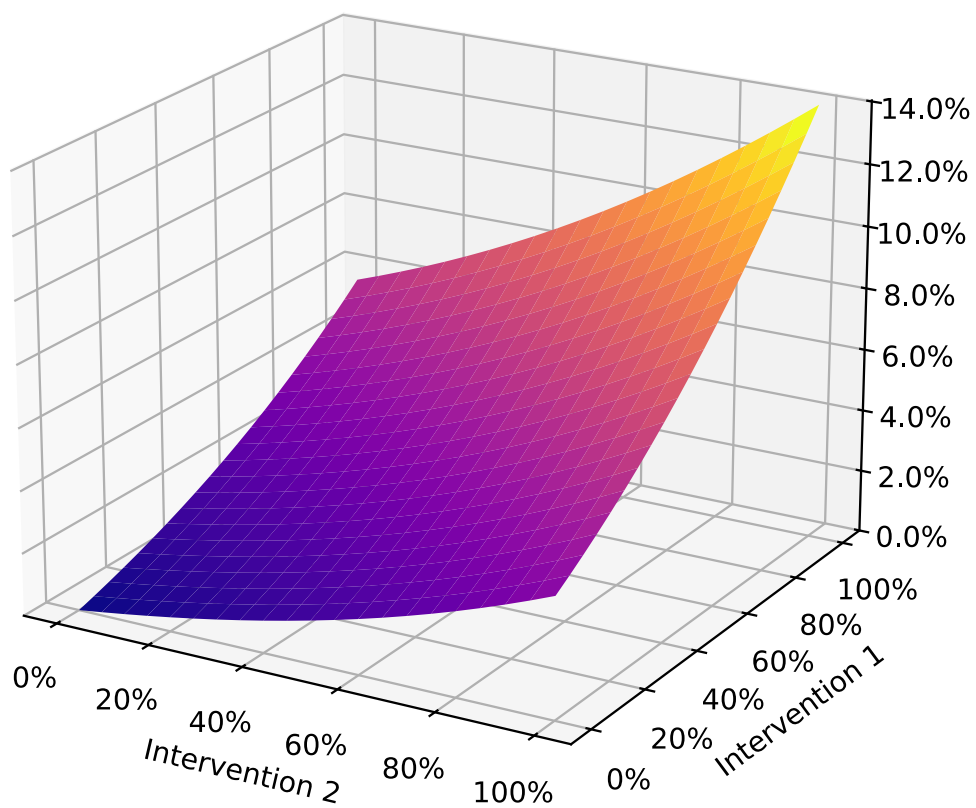


Application of Discrete Choice Models in Policy Impact Analysis

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Notes in regards to the report:

All figures are vectorised, so feel free to zoom in as you please.

1 Introduction and Conceptual Approach

1.1 Background

Public transport provides an equitable means of travel for people of all backgrounds, and is the backbone of many cities, shuttling commuters to work and back. However, it does not have the means to drop all its users directly at their destination, there is always a "first/last mile" that concerns the user, and can be the deciding factor of whether or not they choose to use this mode of transport. The user may even have to drive to the train station just so they can get to work, which mitigates the environmental benefit of public transport. For sustainable urban mobility, the use of public transport and other active methods should be encouraged, to keep congestion down and increase air quality. The micro-mobility global market is set to rise from £37 billion in 2020 to £169 billion in 2030, and shows no sign of slowing down since COVID-19 ([Local Government Association 2023](#)). With the recent rise of micro-mobility, through services such as Lime and Forest providing means of travel through bike and scooter sharing, the "first/last mile" dilemma can be mitigated through similar means.

The main objective of this analysis is to promote sustainable urban mobility by encouraging more people to use public transport. The key motivations are as follows:

- **Reducing Emissions:** encouraging users to use public transport as the use of private vehicles is much more carbon-intensive.
- **Promoting inclusivity and equity:** providing a cheap means of access to public transportation to those who live in rural areas and cannot drive, as hiring private vehicles such as taxis is expensive. Ensuring that everyone has the option to use public transport, and that the last/first mile isn't an issue.

1.2 Stakeholders

To successfully implement policy around micro-mobility based solutions to access to public transport, it is vital that the opinions of internal and external stakeholders are considered. A solution may very well be analytically lucrative, however the key to success of policy and large projects is the agreement of stakeholders towards them; nobody will use a service that they do not like. Table 1 gives a list of potential stakeholders to the intervention implementation.

Table 1: Stakeholders

Stakeholder Types	Examples	Primary interests
Primary Users	- Cost-sensitive commuters, such as seniors, students, and low-income individuals. - Time-sensitive commuters, such as business professionals.	That access to public transport is affordable and reliable, and that their commute time is reduced.
Local Community	- Local residents around where the implementation of the interventions takes place.	That access to public transport is improved in the area, and that micro-mobility services do not create clutter or invite unwanted disturbance in the neighborhood due to noise.
Public Sector	- Local government and public transport agencies.	Promote sustainable, equitable, and potentially profitable access to public transportation services, which themselves generate revenue for public transport agencies and governments.
Private Sector	- Companies such as Lime and Forest.	That the introduction of new services attracts new customers and generates profit.
NGOs	- Environmental advocates such as Greenpeace. - Charities such as Age UK that advocate for different demographics.	That the interventions do not have an adverse environmental effect, and that vulnerable groups receive equitable treatment and consideration when implementing policy.

1.3 Interventions

Providing an easily accessible means of travel to public transport would help increase its use, and for this specific report, the use of micro-mobility will be the primary solution used. This will be investigated using 2 interventions:

1. **Intervention 1:** Slower but free micro-mobility - this has a direct effect on the *access cost* of public transport by reducing it or getting rid of it completely.
2. **Intervention 2:** Faster but more expensive micro-mobility - this has a direct affect on the *access time* of public transport, which implies that the micro-mobility methods are more efficient and direct.

These interventions are applied using discrete choice modelling (DCM). The mode choices considered are car, public transport, walking and cycling. Alongside this, the impact of these interventions will be assessed across different genders and age groups. Note that with the analysis taken, it is assumed that the interventions only affect the utility of public transport - realistically,

this would not be the case, as the introduction of micro-mobility services also introduces a new mode of transport, that being micro-vehicles themselves, however this is not considered.

1.4 Feasibility

1.4.1 Regulation

As intervention 2 is revenue generating, private companies are more likely to take it on, which requires them to adhere to certain rules that require anti-clutter and anti-disturbance measures are taken to ensure stakeholders are satisfied. Many case studies show that micro-mobility is appreciated, although it can become a nuisance if unregulated (Brencher 2024). The feasibility of both interventions is affected by regulations around safe driving, so frameworks that limit speed and age of the user, and encourage helmet-wearing should be put in place to guarantee the longevity of the interventions.

1.4.2 Finances

Intervention 1 prioritises affordability, so it is unlikely any significant revenue is generated; this implies that it would require governmental subsidy to become feasible. This is possible, and has been done successfully many times in the past, with schemes such as Santander bikes in London (Transport for London 2023) and Citi bikes in New York (Flegenheimer 2014), where local governments partnered up with financial institutions to implement bike-share services. Finances should be less of an issue for intervention 2, as it is more likely to generate revenue; consequently, this would be burdensome to low-income individuals.

1.4.3 Infrastructure

One advantage of micro-mobility is that it utilises already existing modes of transport that use road networks, such as e-scooters regular bikes and e-bikes. This makes the upfront cost of setting up micro-mobility hubs to be very minimal. The addition of speed limits allows for use in bike superhighways, which even give access to areas where the car-centric road network cannot. It is important to note that setting up electric micro-vehicles would have a higher upfront cost due to charging docks that must be installed, however this is still much more doable than extending public transport routes.

1.4.4 Maintenance and Security

One concern when implementing the interventions is that micro-vehicles may be stolen or vandalised. This has been a rising issue with Lime bikes in London (Houghton 2023) and there have also been incidents where people would bring their e-bikes into their homes to "preclaim" a bike for their personal commute in the morning. Applying measures to protect against bad actors before and during implementation is paramount. In addition, maintenance is difficult as it would be increasingly hard to recover micro-vehicles that are scattered everywhere, turning maintenance into an overly complex capacitated vehicle routing problem. A balance must be struck between providing commuters complete autonomy with their journey and authorising certain drop-off points that streamline maintenance.

2 Discrete Choice Model Design

2.1 Understanding the dataset

Investigative analysis of the current dataset is important in ensuring that suitable assumptions and decisions are being made with regards to the model. Figure 1 shows that gender and age are represented quite evenly in the sample, although older populations represent slightly smaller proportions. From the mode choice distribution, it can be inferred that the population the sample represents lives in an area with limited links to public transport, whether that be geographically or due to frequency of access. The high percentage of walking and cycling indicates that the average commuter from that population does not necessarily travel far on average, so the distance from home to work isn't the issue stopping commuters from using public transport. It can be deduced that the area represented is a smaller town/borough, where people often travel locally, with some car journeys needed, however more information is needed to be completely sure.

Dataset Statistics

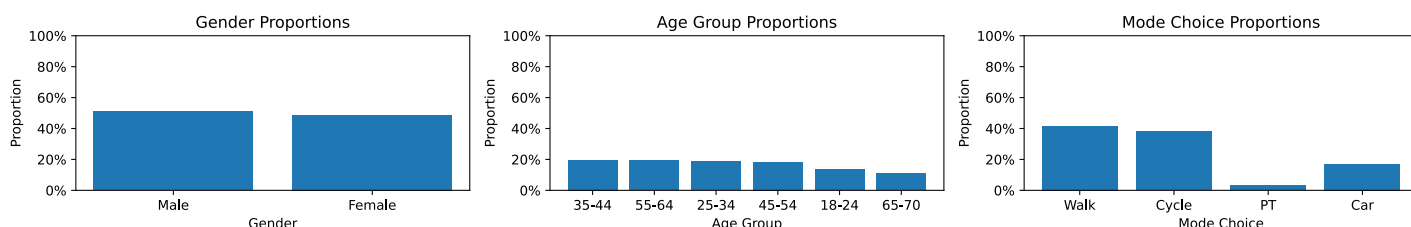


Figure 1: Surface plots for each case, with the maximum value over time

2.2 What type of model?

For this analysis, the MNL was chosen for all alternative models. This decision was made due to time constraints of the project. MNL models still can provide robust estimates of choice probabilities, given that all parameters have low covariance and that all mode choices are independent and their errors are identically distributed. For the dataset given, there was one potential nest that could have been implemented, which is a nest of the active modes of cycling and walking, as it can be

deduced that they are correlated due to shared unobserved factors. This can be done to improve the analysis, as well as the implementation of mixed logit models. One great advantage that MNL models hold is that they provide a closed-form solution, making them extremely simple to compute compared to mixed logit models as well as NL models (although the reasoning is different for NL models - it is because the nests provide another layer of computational complexity). Additionally, the datasets required for the more complex models require more detail, which the current dataset lacks in terms of alternate modes and possibly missing attributes.

2.3 Models considered

To start, the user-related utility that takes into account user-specific attributes is considered as follows in Equation 1:

$$V_{user} = \beta_{age}age + \beta_{gender}gender + \beta_{income}income + \beta_{carownership}carOwnership \quad (1)$$

The main focus of this discussion will be age and gender, although for completeness income and car ownership were included in the user attributes. For this analysis, 3 variations of the MNL model were considered:

Restricted model:

$$\begin{aligned} V_{car} &= \beta_{0,car} + \beta^T TT_{car} + \beta^{TC}(FC_{car} + PC_{car} + CC_{car}) + V_{user} \\ V_{PT} &= \beta_{0,PT} + \beta^T(TT_{PT} + AT_{PT} + WT_{PT}) + \beta^{TC}(AC_{PT} + Fare_{PT}) + V_{user} \\ V_{cycle} &= \beta_{0,cycle} + \beta^T TT_{cycle} + \beta_{cycle}^{bikeability} bikeability_{cycle} + V_{user} \\ V_{walk} &= \beta^T TT_{walk} + \beta_{walk}^{walkability} walkability_{walk} + V_{user} \end{aligned} \quad (2)$$

Model 2 assumes that the change in utility dependent on travel time and travel cost is the same across modes. This is the most restrictive model, as it only contains 11 parameters. All time-related and cost-related factors in the utility functions have been grouped together, as they are quantities of similar nature.

Partially restricted model:

$$\begin{aligned} V_{car} &= \beta_{0,car} + \beta_{car}^T TT_{car} + \beta_{car}^{TC}(FC_{car} + PC_{car} + CC_{car}) + V_{user} \\ V_{PT} &= \beta_{0,PT} + \beta_{PT}^T(TT_{PT} + AT_{PT} + WT_{PT}) + \beta_{PT}^{TC}(AC_{PT} + Fare_{PT}) + V_{user} \\ V_{cycle} &= \beta_{0,cycle} + \beta_{cycle}^T TT_{cycle} + \beta_{cycle}^{bikeability} bikeability_{cycle} + V_{user} \\ V_{walk} &= \beta_{walk}^T TT_{walk} + \beta_{walk}^{walkability} walkability_{walk} + V_{user} \end{aligned} \quad (3)$$

Model 3 frees some of the restrictions and introduces 3 other parameters into the model. The time traveled and the total costs of each mode are assumed to be felt differently through introduction of alternate specific parameters in place of β^T for all modes, and β^{TC} for public transport and car modes. Note that all the time and costs experienced by the user are still grouped together respectively within the utility functions.

Unrestricted model:

$$\begin{aligned} V_{car} &= \beta_{0,car} + \beta_{car}^{TT} TT_{car} + \beta_{car}^{FC} FC_{car} + \beta_{car}^{PC} PC_{car} + \beta_{car}^{CC} CC_{car} + V_{user} \\ V_{PT} &= \beta_{0,PT} + \beta_{PT}^{TT} TT_{PT} + \beta_{PT}^{AT} AT_{PT} + \beta_{PT}^{AC} AC_{PT} + \beta_{PT}^{WT} WT_{PT} + \beta_{PT}^{Fare} Fare_{PT} + V_{user} \\ V_{cycle} &= \beta_{0,cycle} + \beta_{cycle}^{TT} TT_{cycle} + \beta_{cycle}^{bikeability} bikeability_{cycle} + V_{user} \\ V_{walk} &= \beta_{walk}^{TT} TT_{walk} + \beta_{walk}^{walkability} walkability_{walk} + V_{user} \end{aligned} \quad (4)$$

Finally, Model 4 provides alternate specific parameters for all terms, increasing the number of parameters to 20. It is important to note that increasing the number of parameters in a model does not necessarily mean that the model will improve significantly. There comes a point where there may be overfitting in the model, as well as a steep performance tradeoff. To see if releasing some of the restrictions from the models is significant enough, a Likelihood ratio test, alongside BIC values were considered.

2.4 Testing model fit

The likelihood ratio test statistic is χ^2 distributed, and can be calculated through the following equation:

$$-2(\log(L_{Restricted}(\beta)) - \log(L_{General}(\beta))) \quad (5)$$

it can be shown that the degrees of freedom for the χ^2 distribution can be calculated as the difference in the number of parameters. The null hypothesis states that freeing restrictions in the model does not provide a significant enough improvement in this case, whilst the alternative hypothesis states otherwise. For this test, a confidence interval of 95% is taken. In addition to this, the Bayesian Information Criterion (BIC) of each model was considered. It can be calculated using this equation:

$$-2\log(L(\beta)) + K \ln(N) \quad (6)$$

where K is the number of parameters, and N is the number of observations in the dataset (Bierlaire 2023). BIC serves as a good means of measuring the fit of a wide variety of model types, as it punishes excessive parameter use proportional to the observations as well as poor model fit. This methodology allows you to also test non-nested models so it is useful to evaluate. It is important to note that BIC's values are not necessary for judging fit, but rather their differences. It is generally considered that a difference (reduction) in BIC over 10 provides significant improvement between 2 models (Lorah & Womack 2019). Table 2.4 shows the result of the testing conducted on the 3 models.

Table 2: BIC and Likelihood ratio test results.

Values	Restricted model	Partially Restricted model	unrestricted model
BIC	16684.784	16299.477	16076.802
Log Likelihood	-8291.735	-8050.266	-7846.297
No. of Parameters	11	14	20
Test Parameters	Threshold statistic = 7.815		Threshold statistic = 12.591
	Test statistic = 482.938		Test statistic = 207.938
LR Test Results	Null hypothesis rejected		Null hypothesis rejected

From these results, it is quite clear that *the unrestricted model*, model 4 provides the best representative fit to the dataset out of the 3, as it produces significant hypothesis test results in a 95% confidence interval, and provides a reduction in BIC which is far greater than 10 against the other 2 models.

3 Estimation, Application and Results

3.1 Model Parameters

Table 3 contains the estimated parameters calculated in the chosen model. For each parameter, a robust t-test is conducted to see whether or not they have a significant impact when added to the model, compared to where they aren't added at all. According to the p-values produced for the alternate specific constant for cycling, and parameters related to gender and car ownership being over 0.05, these attributes do not provide any significant change to model fit, which implies that they do not affect choice probabilities. This is an important realisation, as the scope of this investigation specifically focuses on the differing impact on gender groups, which can be inferred to be identical to each other. For the sake of the exercise, these parameters have been left in the model, however, it is important to acknowledge that they do not provide any statistically significant effect to the model for a 95% confidence interval. One interesting factor that is statistically significant is income. The impact of income is quite relevant to the interventions, as it would concern whether or not a user can afford them - however, *this user attribute will not be investigated within the report as it is out of scope*.

3.2 Implementation of interventions

To simulate the interventions onto the dataset, a few modifications need to be made to the chosen model. This is available in Equation 7, where the public transport utility is amended. Parameter α represents the percentage reduction in **access time** experienced per user, and γ represents the percentage reduction in **access costs** experienced per user; these added parameters have been implemented for interventions 2 and 1 respectively. 400 different scenarios were simulated, including scenarios that combine elements of interventions 1 and 2 - for example, a scenario where a 20% reduction in access cost and a 50% reduction in access time to public transport is simulated. Note that the interventions only affect the utility of public transport, and that although intervention 1 considers free (100% reduction) access costs, other reduction values in access cost are still considered.

Intervention Simulation:

$$V_{PT} = \beta_{0,PT} + \beta_{PT}^{TT} TT_{PT} + (1 - \alpha)\beta_{PT}^{AT} AT_{PT} + (1 - \gamma)\beta_{PT}^{AC} AC_{PT} + \beta_{PT}^{WT} WT_{PT} + \beta_{PT}^{Fare} Fare_{PT} + V_{user} \quad (7)$$

3.3 Impacts on Utility

In figure 2, as time traveled increases for the average user, the utility of public transport also increases. This is also the case for both increasing access cost and access time. Peculiarly, it is observed that as reductions increase for both access time and access costs, the utility decreases - this was further investigated, and as seen in figures 6 and 7 in Appendix A: Utility functions, and it appears that this also occurs in utilities in other modes, so it may occur due unobserved attributes in all

Table 3: Summary of all the parameters in the chosen model

Parameter	Value	Robust Standard error	Robust t-test	Robust t-test p-value
$\beta_{0,PT}$	3.36	0.326	10.3	0.0
$\beta_{0,car}$	3.04	0.172	17.7	0.0
$\beta_{0,cycle}$	0.0644	0.0985	0.653	0.514
β_{PT}^{AC}	-0.567	0.0444	-12.8	0.0
β_{PT}^{AT}	-0.097	0.00942	-10.3	0.0
β_{Age}	-7.38e-14	1.96e-14	-3.76	0.000168
$\beta_{cycle}^{Bikeability}$	0.00536	0.000861	6.22	5.07e-10
$\beta_{CarOwnership}$	9.52e-17	5.3e-15	0.018	0.986
β_{CC}^{car}	-0.129	0.0129	-10.0	0.0
β_{PT}^{Fare}	-1.14	0.069	-16.5	0.0
β_{car}^{FC}	-0.553	0.0399	-13.9	0.0
β_{Gender}	3.75e-16	3.84e-16	0.977	0.329
β_{Income}	7.86e-14	1.27e-15	61.9	0.0
β_{car}^{PC}	-0.243	0.0146	-16.7	0.0
β_{PT}^{TT}	-0.0691	0.00558	-12.4	0.0
β_{car}^{TT}	-0.106	0.00299	-35.5	0.0
β_{cycle}^{TT}	-0.0722	0.00184	-39.3	0.0
β_{walk}^{TT}	-0.0701	0.00176	-39.8	0.0
β_{PT}^{WT}	-0.0451	0.0129	-3.49	0.00049
$\beta_{walk}^{Walkability}$	0.00808	0.000835	9.68	0.0

of the modes. Note the nonlinear increase in utility with the increase in access costs in intervention 1 and access times in intervention 2, implies that the reductions apply an increasing incentive to those who experience high access costs and times.

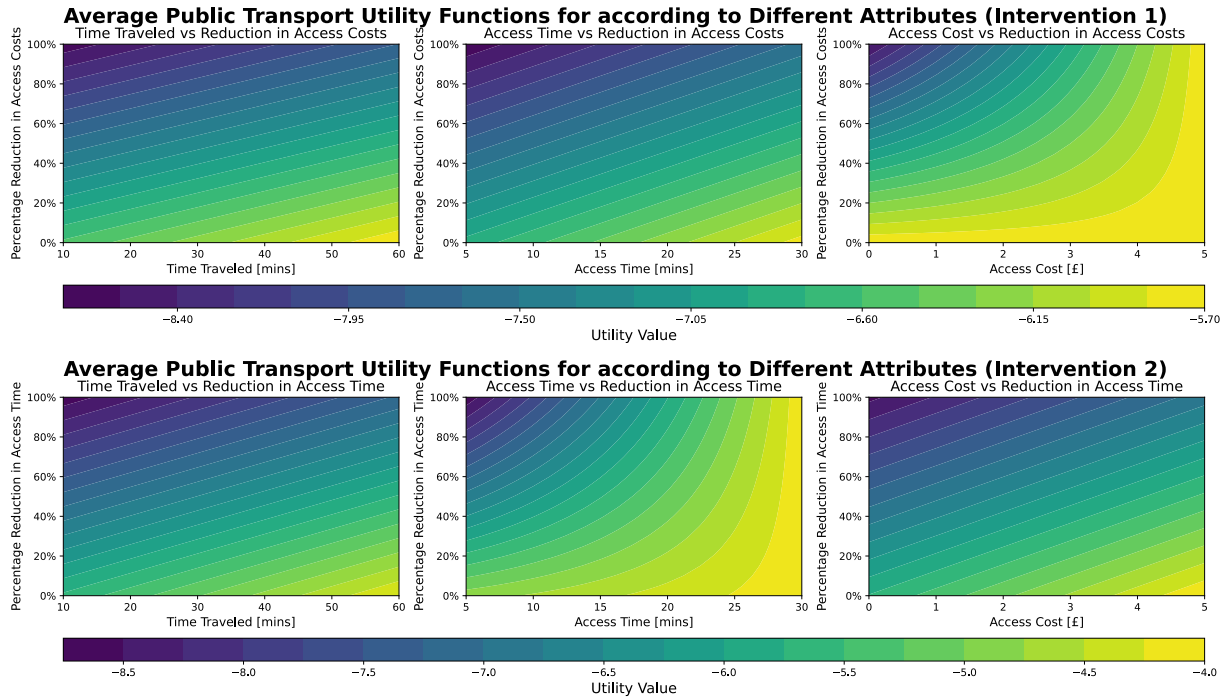


Figure 2: Utility experienced according to travel time, access cost and access time for interventions

3.4 Results

Figure 3 shows the resulting change in market share, which was calculated using the sample enumeration method. It is clear that as the intensity of implementation of both interventions increases, the predicted market share of public transport also increases. This has an adverse effect on the other mode choices, especially cycling and walking.

Intervention 1, where a 100% reduction in access cost is applied, adds 3.865% to the public transport market share, while it takes away 1.666%, 1.534% and 0.665% away from the walking, cycling and car market shares respectively. To achieve a similar market share increase through only applying intervention 2, an 80% reduction in access time is required. This amounts to an added 3.990% market share to public transport which is taken from 1.733%, 1.544% and 0.713% from walking, cycling and car market shares. If the interventions are considered exclusively, then it would be *much more feasible to implement intervention 1*, as achieving an 80% reduction (see figure 8 in [Appendix B: Market shares](#)) in access time is idealistic - many factors are unchangeable such as distances from stations and road speed limits that would make an 80% reduction very hard to achieve. If however both interventions can be used in conjunction, improvements to public market share can happen

Change in Total Market Shares for Interventions

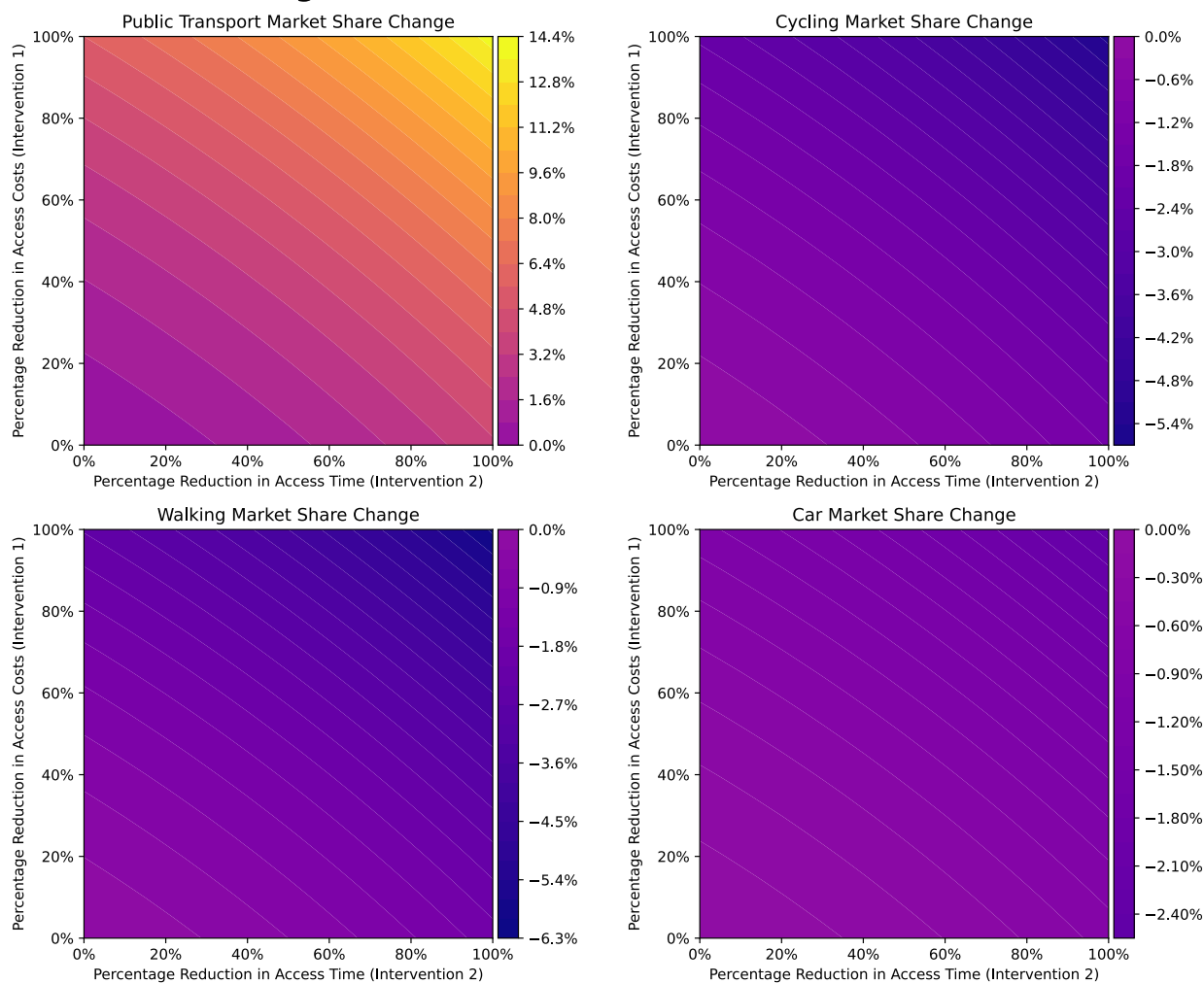


Figure 3: Change in Market Shares from Base Case

at lower access time reductions. For example, if a 50% cost reduction and a 50% time reduction in access is instated, then a 4.299% addition to public market share can be achieved.

Public Transport Market Share for Different Genders

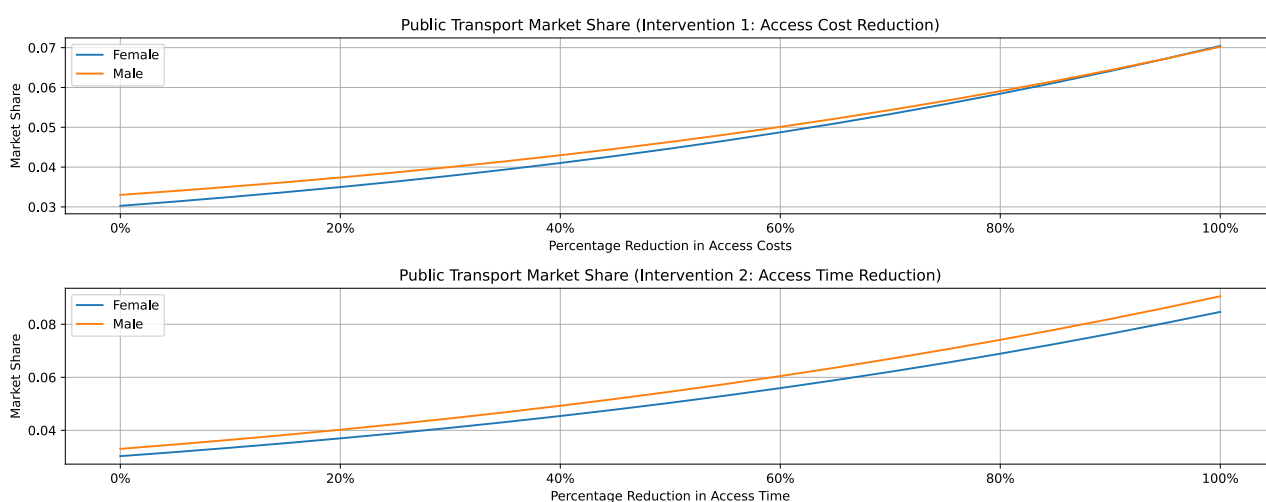


Figure 4: Effects of Age on Market Share

Figures 4 and 5 show the effects of the interventions on the groups of interest. It was found that the gender parameter is not statistically significant, however, figure 4 implies that females are more responsive to reductions in access cost - this is likely due to other factors that are coincidentally shared amongst women such as income due to the previous t-test. Access time reduction provides similar results across genders as expected. Access cost reduction has the highest change in market share for age groups between 25-34 and 35-44 as per figure 5, whilst access time reduction impacts all age groups similarly. This

implies that middle-aged users are more concerned about the cost of accessing public transport whilst all age groups value access reduction times equally. See [Appendix B: Market shares](#) for the surface plots of age and gender.

Public Transport Market Share for Different Age Groups

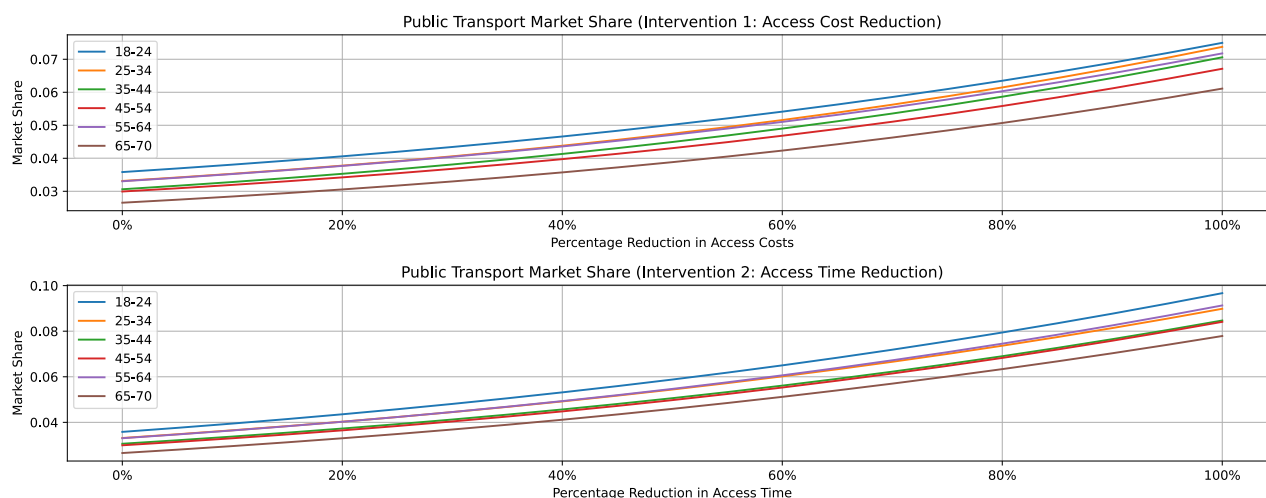


Figure 5: Effects of Gender on Market Share

4 Discussion and Conclusions

It is clear that both interventions increase the public transportation market share, which from a profitability standpoint is positive, as they drive new sources of revenue for public transport authorities from other mode choices. However, since most of the increase comes from people who have previously walked or cycled to commute (active modes), that begs the question - would improving public transport usage with micro-mobility adversely affect the region it represents? It seems that the age groups that are usually deemed as low-income (young people, seniors) aren't as enticed to use public transport as middle-aged persons through these interventions, and that car users are not as attracted to these interventions in general. Micro-vehicles do not tend to generate any direct emissions, however, contribute to an increased use of electricity, an increase in e-waste and an increase in emissions due to higher congestion (which would decrease the area's air quality). Moreover, it was previously determined that combining aspects of both interventions actually produces a higher impact with lower reductions, which should definitely be considered. Even then, the combination that produces the best result may not be the most suitable, there are factors such as budget and stakeholder satisfaction to account for. It is vital for policymakers to acknowledge the bigger picture beyond the objectives of the interventions that they put in place.

On the other hand, the results obtained from the analysis may be deemed to be inaccurate due to the methodology. The use of MNL models assume that all modes and attributes are uncorrelated and that the associated errors are independent and identically distributed, which is unrealistic. Another issue is that the data used is static, and the model assumes that the parameters are constant due to the data provided. This is clearly an issue as travel patterns can be seasonal, so to improve the demand modelling, it would be wise to include a temporal aspect. There are also limitations with the attributes that may be missing from the dataset, such as the purpose of the journey, which may affect the taste parameters of other attributes. For example, a person travelling leisurely might view the same journey as more expensive if they were commuting instead. The model also considers that all attributes are linear and linearly related, this can be inaccurate, as the interactions between attributes present a complex relationship. It is important to note that this model is only fitted to the population that the sample represents, so impacts are not necessarily translatable across populations, which makes the model inflexible. In addition, the model is incapable of anticipating unexpected travel events such as large concert crowds and would fail to predict how people would travel. Finally, only first-order travel flows and single-mode journeys are considered with this model, which may produce inaccurate demand patterns within the sample.

To improve model fit, statistical tests should be conducted between different model types such as NL and MxL models to see which model provides the most accurate prediction of choice probabilities. Finally, the model does not consider using micro-mobility as a mode choice itself, although it has theoretically been implemented; this suggests that people are not going to use it as a full means of travel, which realistically will happen if introduced. This might make the change in market share less accurate, so modelling the introduction of an additional mode choice would improve accuracy; this would require research on the attributes of the micro-mobility mode choice, so the current dataset cannot handle this implementation.

In conclusion, both interventions provide an increase in public transport mode share, at the expense of a majority decrease in walking and cycling mode shares. Policymakers should consider a combination of both interventions to achieve a greater mode shift with smaller reductions. They should also consider the environmental impact of the increased vehicle mode share in relation to increase in congestion and decrease in active mode shares. Finally, policymakers should account for stakeholder satisfaction, to ensure that the interventions are received well within the community.

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4.1 Appendix A: Utility functions

Utility Functions for according to Different Modes (Intervention 1)

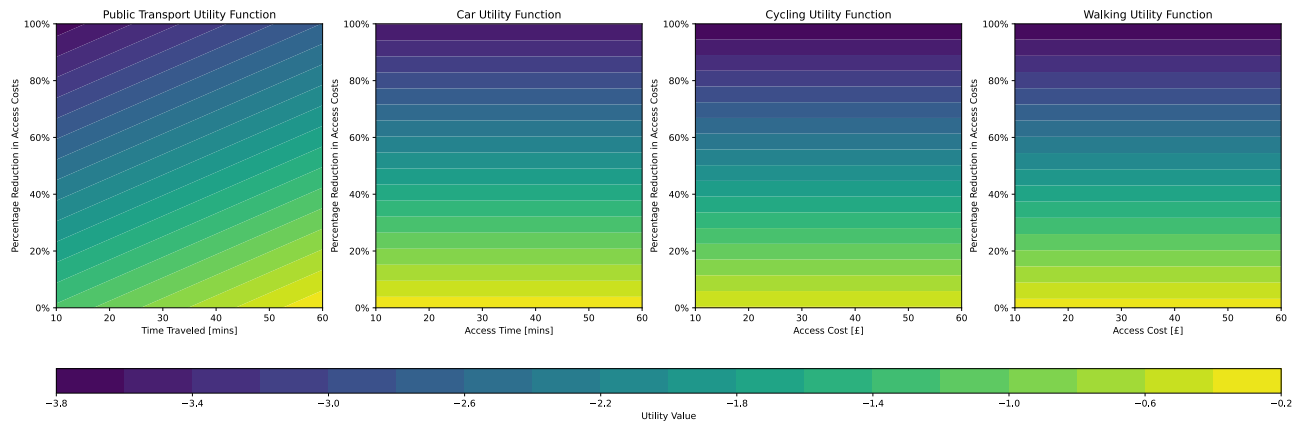


Figure 6: Utility experienced for different modes for Intervention 1

Utility Functions for according to Different Modes (Intervention 2)

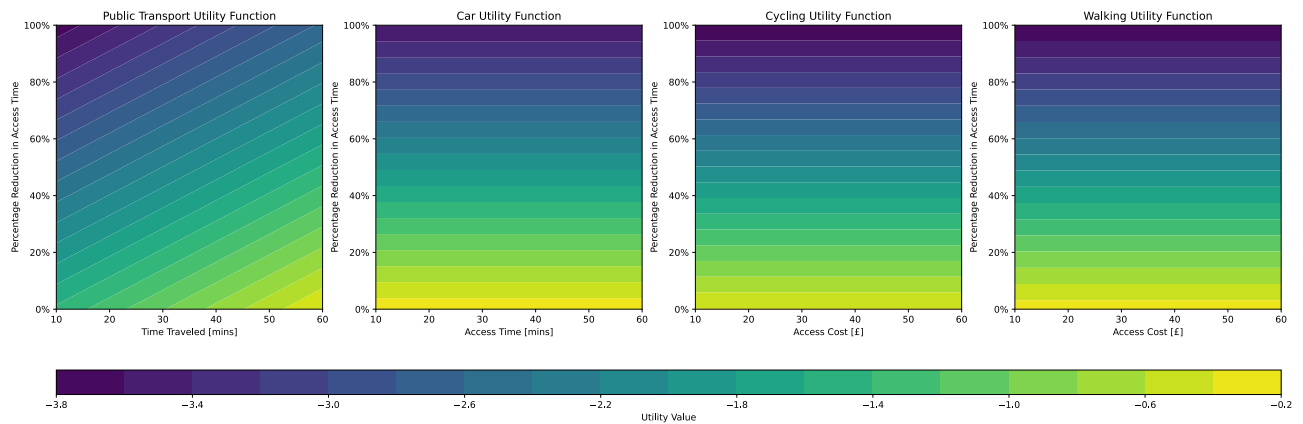


Figure 7: Utility experienced for different modes for Intervention 2

4.2 Appendix B: Market shares

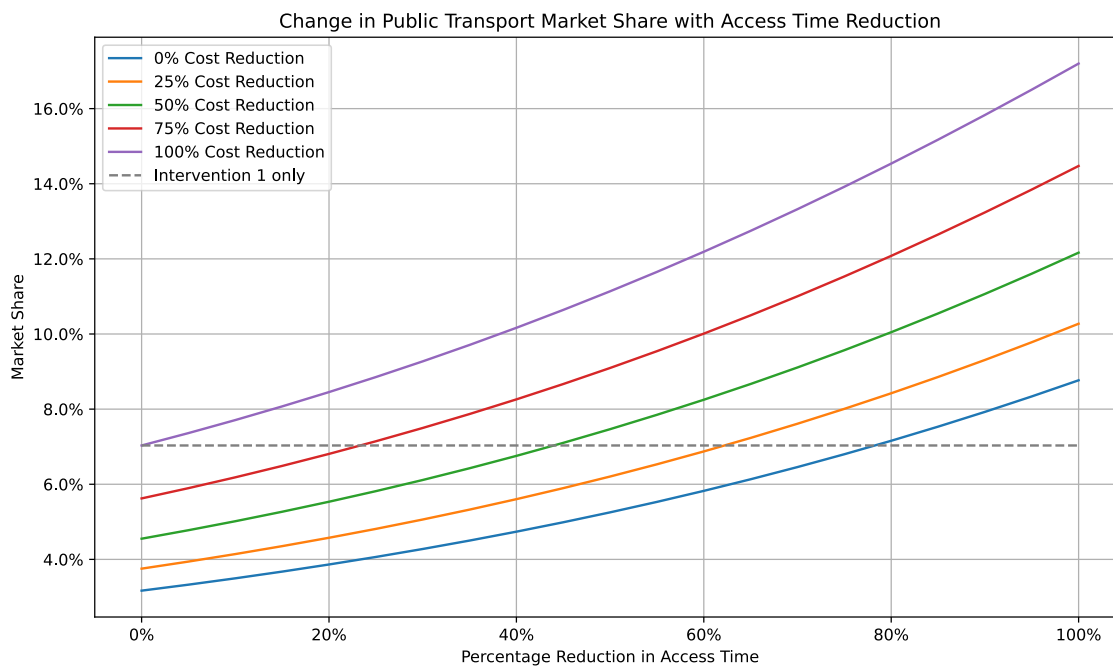


Figure 8: Public Transport Market share lines against Access Time Reduction

Public Transport Market Shares by Age Group

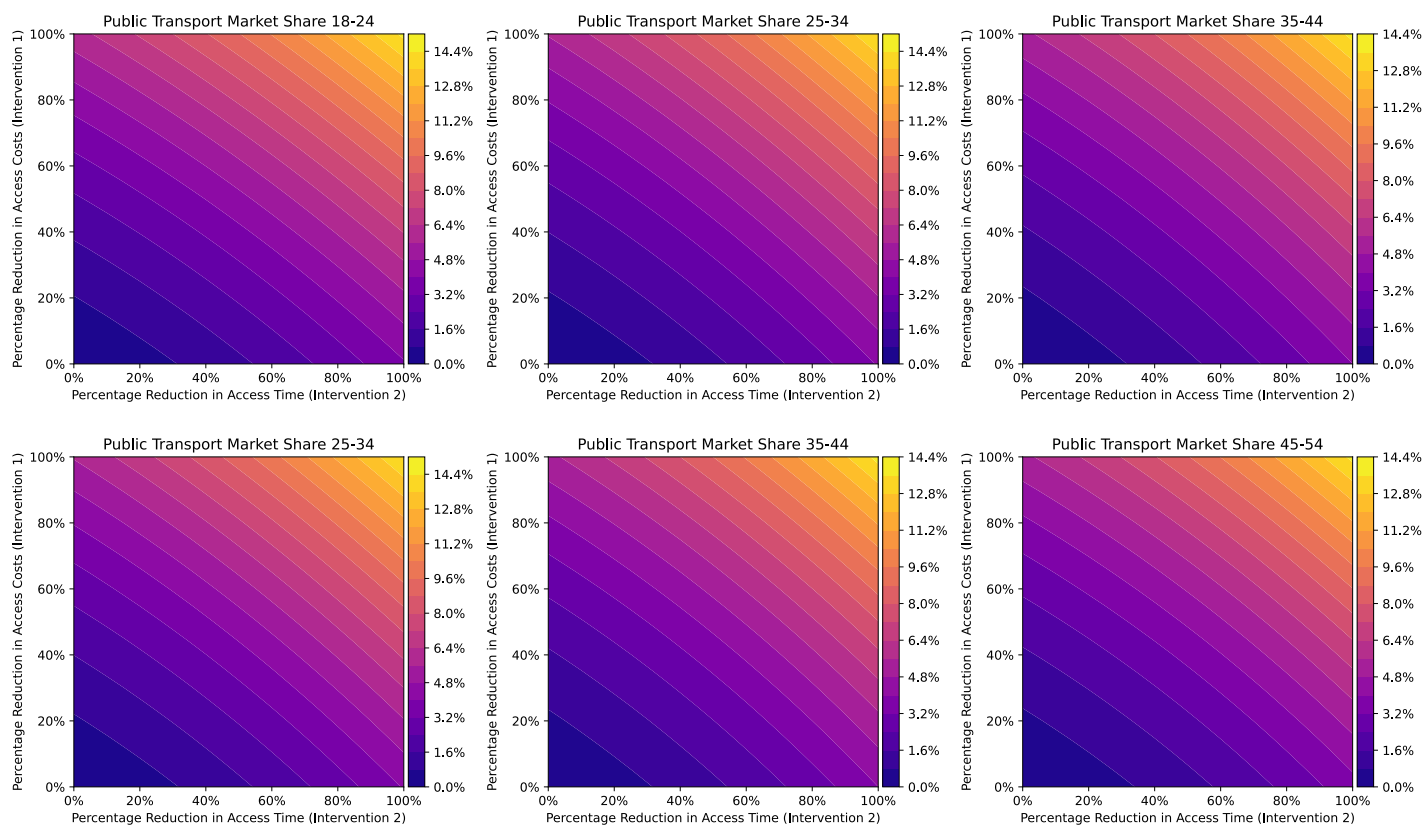


Figure 9: Public Transport Market Share for each Age Group

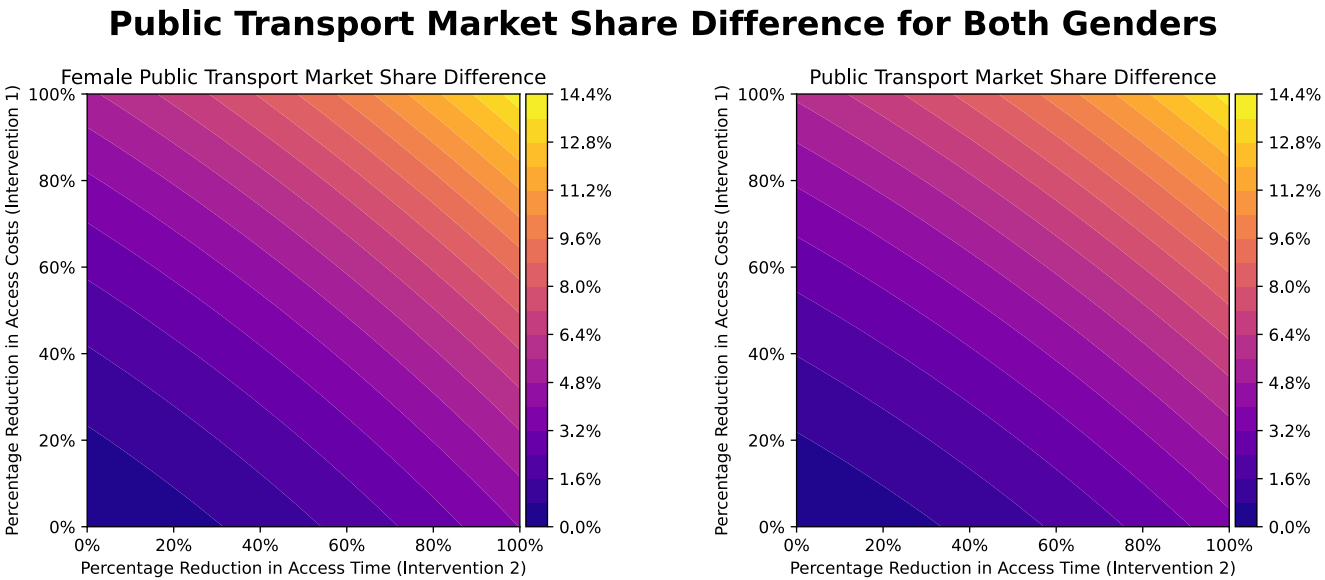


Figure 10: Public Transport Market Share for each Gender